Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

LESSON PLAN

For

Data Analytics and Visualization with Power BI

TGS Ref No: TGS-2020505444

Conducted by

Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

Version 15.0

DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
14.0	01 July 2024	Legacy Power BI courseware and exercise set.	Tertiary Infotech
15.0	20 August 2026	Rebuilt visual deck; aligned detailed Learner Guide, slide-mapped Lesson Plan, ten self-contained labs, editable PowerPoint charts, current WSQ administration and assessment mapping.	Dr Alfred Ang

TABLE OF CONTENTS

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Open this document in Microsoft Word and update the table if prompted.

Course Overview

A two-day, 16-hour WSQ course in which learners prepare data, design analytical reports, build dashboards, create interactive experiences, model a star schema and analyze data with DAX. Procedures are practiced in ten progressive labs and assessed through WA and PP.

Learning Outcomes

- LO1: Perform data transformation techniques using Power BI
- LO2: Create data visualizations that portray patterns, trends and correlations
- LO3: Develop dashboards and scorecards for decision monitoring
- LO4: Create dynamic and interactive data visualizations
- LO5: Create a data model for strategic visualization and mapping
- LO6: Perform data analytics and queries with DAX

Daily Schedule

Time	Duration	Topic / Activity	Method	Slides
Day 1 · 9:00–9:30	Varied	Administration, outcomes and assessment briefing	Administration / assessment	Slides 1–16
Day 1 · 9:30–12:45	Varied	Topic 1 and Labs 1–2: data intake and transformation	Facilitated visual explanation + guided lab	Slides 17–38
Day 1 · 12:45–1:30	—	Lunch	Administration / assessment	—
Day 1 · 1:30–3:20	Varied	Topic 2 and Labs 3–4: visual design and report story	Facilitated visual explanation + guided lab	Slides 39–60
Day 1 · 3:20–3:30	—	Break	Administration / assessment	—
Day 1 · 3:30–5:00	Varied	Topic 3 and Lab 5: dashboards and scorecards	Facilitated visual explanation + guided lab	Slides 61–78
Day 1 · 5:00–6:00	Varied	Applied review, evidence check and reflection	Administration / assessment	Slides 61–78
Day 2 · 9:00–11:10	Varied	Topic 4 and Lab 6: interactions and navigation	Facilitated visual explanation + guided lab	Slides 80–97
Day 2 · 11:10–11:20	—	Break	Administration / assessment	—
Day 2 · 11:20–12:45	Varied	Topic 5 and Lab 7:	Facilitated visual	Slides 98–116

Time	Duration	Topic / Activity	Method	Slides
		semantic modeling	explanation + guided lab	
Day 2 · 12:45–1:30	—	Lunch	Administration / assessment	—
Day 2 · 1:30–3:20	Varied	Topic 6 and Labs 8–10: DAX, analytics and capstone	Facilitated visual explanation + guided lab	Slides 117–141
Day 2 · 3:20–3:30	—	Break	Administration / assessment	—
Day 2 · 3:30–6:00	Varied	Assessment: WA 60 min + PP 90 min	Administration / assessment	—

Topic-by-Topic Delivery

Topic 1: Get Started with Power BI and Power Query · Slides 17–38

Alignment: K1 · A1 · LO1. Facilitate concepts through the visual frameworks and native charts, then shift procedures to the lab guide/Learner Guide.

- From data to action: Descriptive, diagnostic, predictive, prescriptive and cognitive analytics support different decisions.
- Roles in the data team: Business experts frame the question; analysts transform evidence; engineers govern dependable data flows.
- The analyst workflow: Prepare → model → visualize → analyze → publish → monitor; each stage leaves evidence for review.
- Power BI ecosystem: Desktop authors reports and semantic models; Service distributes, refreshes, governs and collaborates.
- Dataset, semantic model, report, dashboard: Separate reusable data logic from multi-page analysis and single-page monitoring.
- Desktop views: Report view composes visuals, Data view inspects loaded tables, and Model view manages relationships.
- Connection choices: Files, folders, databases, web sources and cloud services differ in refresh, scale and governance.
- Import or DirectQuery: Import favors speed and modeling freedom; DirectQuery favors source currency but adds dependency and latency.
- Navigator decisions: Select only required tables and preview structure before loading to reduce noise and model size.
- Power Query Editor: Applied Steps make transformations visible, ordered, repeatable and auditable.
- Promote headers and remove rows: Fix structural metadata before analytical logic so column names and data boundaries are reliable.
- Choose and rename columns: Keep only decision-relevant fields and give them business-friendly names.
- Pivot and unpivot: Wide layouts are human-friendly; long tidy layouts are usually better for filtering, aggregation and visuals.
- Replace values, nulls and errors: Differentiate genuine zero, missing, not applicable and invalid data before replacement.

- Remove duplicates: Define the business key before deduplication; repeated rows are not always duplicates.
- Data types and locale: Types determine aggregation and date behavior; locale determines how text values are interpreted.
- Append queries: Stack tables with the same grain and schema, then reconcile before/after row counts.
- Merge queries: Join tables by validated keys; left outer, inner and full outer joins answer different integrity questions.
- Column profiling: Quality, distribution and profile views surface nulls, errors, outliers and unexpected cardinality.
- M and Advanced Editor: Power Query records transformations as M; readable steps improve reuse and troubleshooting.

Lab 1: Connect, Inspect and Profile an Excel Dataset · Slide 37

Output: A profiled query with correct headers, types and documented quality issues Acceptance: All required tables load; column quality is reviewed; no unintended errors remain.

Lab 2: Clean, Transform, Append and Merge · Slide 38

Output: A repeatable Power Query transformation pipeline with an audit trail Acceptance: Headers, types, nulls and duplicates are corrected; append and merge row counts reconcile.

Topic 2: Create Data Visualizations and Reports · Slides 39–60

Alignment: K2 · A2 · LO2. Facilitate concepts through the visual frameworks and native charts, then shift procedures to the lab guide/Learner Guide.

- Start with the question: Choose a visual only after naming the comparison, trend, distribution, relationship or composition to reveal.
- Column and bar charts: Use length on a common baseline for accurate category comparison; sort to make ranking visible.
- Line and area charts: Use lines for change over ordered time; use area sparingly when magnitude and composition both matter.
- Cards, KPIs and gauges: Cards state a value, KPIs add direction and targets, and gauges show progress toward one bounded goal.
- Combo charts: Combine columns and a line when two measures share a meaningful category and their scales remain interpretable.
- Decomposition tree: Let users drill across candidate dimensions to explain a result without predefining one path.
- Pie, doughnut and funnel: Use part-to-whole displays for few categories and funnels only for genuinely ordered stages.
- Maps and choropleths: Use symbols for magnitude and filled regions for rates; normalize before comparing unequal geographies.
- Tables and matrices: Tables preserve precise records; matrices add hierarchy, subtotals and cross-tab analysis.
- Scatter and bubble charts: Position reveals correlation, while size and color add variables; association is not causation.
- Ribbon, treemap and waterfall: Show rank change, hierarchical composition and additive bridges when those structures answer the question.
- Q&A and smart narratives: Natural-language and generated summaries accelerate exploration but still require human validation.
- Numeric, grid and performance visuals: Match information density to the decision: glanceable status, precise lookup or target tracking.
- Change and resize visuals: Changing type can change meaning; protect labels, proportions and hierarchy when reflowing a page.

- Format for comprehension: Use consistent color roles, restrained emphasis and direct labels to reduce decoding effort.
- Titles and backgrounds: Insight titles state the takeaway; backgrounds should organize, not decorate.
- Tooltips: Use tooltips for supporting context, definitions and secondary measures without crowding the main canvas.
- Drill down and expand: Hierarchies support movement between levels; keep the current level and path visible.
- Themes: A theme standardizes color, type and defaults so reports remain coherent and maintainable.
- Accessibility: Reading order, alt text, contrast, keyboard focus and non-color cues make insight available to more users.

Lab 3: Choose Visuals for Trends and Correlations · Slide 59

Output: A one-page analytical report using fit-for-purpose native visuals Acceptance: Each visual answers a stated question; encodings are accurate; titles express insight.

Lab 4: Design an Accessible Report Story · Slide 60

Output: A polished report page with hierarchy, theme, tooltips and accessible navigation Acceptance: The page has a clear reading order, consistent formatting, alt text and no misleading scale.

Topic 3: Develop Dashboards and Scorecards · Slides 61–78

Alignment: K3 · A3 · LO3. Facilitate concepts through the visual frameworks and native charts, then shift procedures to the lab guide/Learner Guide.

- Report versus dashboard: Reports are multi-page interactive analysis; dashboards are one-page monitoring surfaces in Power BI Service.
- Dashboard development cycle: Define decisions → select KPIs → design tiles → publish → test → monitor and refine.
- Pin a visual: Pinned tiles create a stable monitoring surface while preserving a path back to the source report.
- Pin a live page: A live page retains report interactions; use it when monitoring needs more than independent tiles.
- Combine multiple reports: A dashboard can assemble governed signals from several reports into one executive view.
- Alerts: Alerts turn threshold breaches into action; they require clear owners, frequency and escalation rules.
- Ask questions: Dashboard Q&A supports fast exploration when model names, synonyms and relationships are business-friendly.
- Quick insights: Automated patterns are hypotheses to review, not conclusions to accept without context.
- Dashboard themes: Consistent themes support recognition across tiles without overwhelming status colors.
- Mobile layout: Prioritize the few signals and actions needed on a small screen; do not merely shrink desktop design.
- Scorecards and goals: A scorecard adds targets, owners, status and check-ins to make performance accountable.
- Workspaces: Workspaces separate authoring, review and distribution responsibilities.
- Apps and sharing: Apps package approved content for audiences; direct sharing is narrower and harder to govern at scale.
- Refresh and gateways: Refresh design connects source availability, credentials, schedules and failure monitoring.

- Dashboard storytelling: Arrange tiles from outcome to drivers to exceptions so executives can scan, diagnose and act.
- Dashboard QA: Verify current data, tile links, mobile view, ownership, permissions and decision usefulness before release.

Lab 5: Build an Executive Dashboard and Scorecard · Slide 78

Output: A Sales Monitoring dashboard and KPI scorecard in Power BI Service Acceptance: Tiles and targets are current, decision-focused and traceable to the report.

Topic 4: Create Dynamic and Interactive Visualizations · Slides 80–97

Alignment: K4 · A4 · LO4. Facilitate concepts through the visual frameworks and native charts, then shift procedures to the lab guide/Learner Guide.

- Interaction model: Selections propagate through relationships and visual interactions; every response should be intentional.
- Filter scope: Dataset security, report filters, page filters, visual filters and drillthrough each control a different boundary.
- Cross-filter: A selection removes unrelated marks so the page focuses on the chosen segment.
- Cross-highlight: A selection highlights contribution while retaining the whole for comparison.
- Edit interactions: Configure each visual to filter, highlight or ignore the source selection based on the stakeholder question.
- Slicers: Expose the few high-value filters users need and show the active selection clearly.
- Sync slicers: Shared filter state creates continuity across pages; avoid hidden states that surprise users.
- Hierarchies: Ordered levels support drill paths such as Year → Quarter → Month or Region → Country → Store.
- Drillthrough: Carry context from a summary mark to a focused detail page with a clear route back.
- Bookmarks: Capture filter, display and navigation states to build guided stories and reset controls.
- Buttons and navigation: Use consistent labels, states and placement so navigation feels predictable.
- Conditional formatting: Use rules to direct attention to exceptions, not to decorate every value.
- Report-page tooltips: Provide rich context on hover while keeping the primary view simple.
- Field parameters: Let users switch dimensions or measures without duplicating pages.
- What-if parameters: Turn a report into a controlled scenario tool while distinguishing assumptions from actuals.
- Interactive UX: Preserve orientation with page titles, breadcrumbs, reset state and visible filter context.
- Stakeholder review: Test the display against real questions, expected actions and common misinterpretations.

Lab 6: Create Dynamic Interactions and Navigation · Slide 97

Output: An interactive report with slicers, drillthrough, bookmarks and tested interactions Acceptance: Every interaction is intentional; drill paths work; reset navigation restores the default state.

Topic 5: Create a Strategic Data Model · Slides 98–116

Alignment: K5 · A5 · LO5. Facilitate concepts through the visual frameworks and native charts, then shift procedures to the lab guide/Learner Guide.

- Semantic model: A model organizes reusable business meaning: tables, relationships, measures, hierarchies and security.
- Star schema: A central fact table records events; surrounding dimensions describe who, what, where and when.

- Fact tables: Facts have a declared grain, foreign keys and numeric observations suitable for aggregation.
- Dimension tables: Dimensions provide stable descriptive attributes for grouping, filtering and labeling.
- Relationship keys: Use unique dimension keys and matching fact foreign keys; validate unmatched and duplicate values.
- Cardinality: One-to-many is the normal star pattern; one-to-one and many-to-many require explicit justification.
- Cross-filter direction: Single direction is easier to reason about; bidirectional filters can create ambiguity and unexpected totals.
- Many-to-many: Bridge tables make shared membership explicit and prevent uncontrolled duplication.
- Date dimension: A contiguous date table enables consistent calendar attributes and time intelligence.
- Table and column properties: Names, formats, data categories and default summarization shape the authoring experience.
- Hierarchies: Curated hierarchies guide valid drill paths and reduce repeated setup.
- Measures: Measures calculate at query time and respond to filter context.
- Calculated columns: Columns calculate row by row at refresh and increase model size; use them only when a stored attribute is needed.
- Granularity: Declare one row's meaning before combining sources or writing measures.
- Role-playing dimensions: One Date table can relate to order, ship and due dates; only one relationship is active at a time.
- Model performance: Remove unused columns, prefer star schemas, reduce high cardinality and keep transformations upstream where practical.
- Business-friendly model: Hide technical keys, group measures and use descriptions so authors find the right fields.
- Model QA: Check uniqueness, referential integrity, cardinality, filter paths, totals and refresh behavior before visualization.

Lab 7: Model a Star Schema and Date Dimension · Slide 116

Output: A semantic model with fact/dimension relationships and a marked date table Acceptance: Relationships use correct keys/cardinality; filters flow predictably; model ambiguity is zero.

Topic 6: Analyze Data with DAX · Slides 117–141

Alignment: A6 · LO6. Facilitate concepts through the visual frameworks and native charts, then shift procedures to the lab guide/Learner Guide.

- DAX purpose: DAX defines reusable calculations over a semantic model; it is not a replacement for data cleaning.
- Formula syntax: A calculation has a name, expression, functions, operators and references to tables, columns or measures.
- Reference rules: Columns require table context; measures should be referenced by measure name to preserve reusable logic.
- Data types and operators: Type compatibility controls comparison, arithmetic and blank handling.
- Row context: Calculated columns and iterators evaluate one row at a time.
- Filter context: Visual axes, slicers, filters and CALCULATE define the subset a measure evaluates.
- CALCULATE: CALCULATE evaluates an expression under modified filter context and is central to analytical DAX.
- Variables: VAR and RETURN improve readability, avoid repeated evaluation and simplify debugging.
- Implicit and explicit measures: Explicit measures centralize definitions, formatting and governance.
- Core measures: Start with additive Sales, Cost and Quantity, then derive Profit and Margin from governed measures.
- Safe division: DIVIDE handles zero or blank denominators more safely than the division operator.

- Share of total: REMOVEFILTERS or ALL can create a denominator while retaining intentional dimensions.
- Date table: A marked date table with continuous dates underpins consistent time calculations.
- Time intelligence: YTD, prior period and growth measures depend on valid date relationships and filter context.
- TOPN: TOPN returns a table ranked by an expression; pair it with clear ties and filter behavior.
- Groups and bins: Grouping creates analytical segments; bins reveal distribution at a chosen resolution.
- Outliers and anomalies: Flag exceptions using statistical context, then investigate causes rather than treating detection as explanation.
- Clustering: Clusters suggest similar records based on selected variables; business review is needed to name and use them.
- Analyze feature: Automated increase/decrease explanations provide candidate drivers that require validation.
- What-if analysis: Disconnected parameter tables let users test assumptions without changing actual data.
- Trend lines and forecasts: Trend and forecast overlays summarize direction and uncertainty; they do not prove causality.
- Align findings with SMEs: Translate measures into a claim, supporting evidence, limitations and an action for domain review.

Lab 8: Create Core DAX Measures · Slide 139

Output: Validated measures for Sales, Cost, Profit, Margin and share of total Acceptance: Measures reconcile to workbook controls and respond correctly to filter context.

Lab 9: Apply Time Intelligence and Scenario Analysis · Slide 140

Output: YTD, prior-period, Top N and what-if analysis with decision commentary Acceptance: Date logic is valid; parameters change results; findings state business meaning and limitations.

Lab 10: Deliver the GlobalTech Sales Performance Cockpit · Slide 141

Output: A stakeholder-ready PBIX report, dashboard evidence and QA checklist Acceptance: The solution meets all six outcomes, answers stakeholder questions and passes the evidence checklist.

Resources Required

- Trainer and learner slide PDFs
- Learner Guide DOCX/PDF
- Ten complete lab folders with MD/PDF guides, Excel datasets and PBIX starters
- Power BI Desktop on Windows
- Power BI Service training workspace
- LMS-TMS access and SSG TRAQOM attendance QR code

Assessment

Assessment runs on Day 2 from 3:30 PM to 6:00 PM: Written Assessment (WA), 60 minutes, followed by Practical Performance (PP), 90 minutes. Both are individual and open book using the slides, Learner Guide and approved materials. Learners complete assessment digital attendance, submit the completed answers on the LMS, receive feedback, and sign the Assessment Summary Record. Competency and funding require the prescribed assessment outcome and at least 75% attendance.

LMS: https://lms-tms.tertiaryinfotech.com/